

ERIC BOITTIER, PH.D.

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ARLESHEIM, BL 4144, SWITZERLAND

May 09, 2026

Prof. Dr. Ece Özkan Elsen

Department of Biomedical Engineering
Universität Basel
Hegenheimermattweg 167C
CH-4123 Allschwil, Switzerland

Dear Prof. Dr. Ece Özkan Elsen,

I am writing to express my interest in the Postdoctoral Researcher position with the Analytics & Informatics for Child Health group at the Basel Research Center for Child Health. I believe I have a lot to offer you and your team, and that this role would be a natural extension of my existing research interests and competencies.

The story behind my early research career started in medicinal chemistry, motivated by the goal of improving the quality of life of patients. I have always had an interest in programming and statistics, and my first research position was done in a multidisciplinary team fighting cancer and geriatric disease. It was the deep learning revolution around 7 years ago that drew me to Basel to begin my doctoral studies applying Graph Neural Networks to physics-based (bio)molecular modelling.

In addition to deep learning, I have also developed kernel-based methods and applied Bayesian optimisation. Uncertainty quantification and explainable AI are areas of overlap between our research. I am interested in digital health problems where machine learning can support clinical research and improve pediatric healthcare. I am interested in learning more about data governance, security, pseudonymization, and collaboration with clinical IT partners.

Working at the intersection of AI/ML and data-intensive research, I have experience with computational workflows, version control, containerized environments, and reproducible research setups. Through the BioZentrum, I have attended courses and seminars on HPC; and I am an experienced user of the SciCore cluster. I would be happy to support the group's IT administration needs, including coordination with university IT and HPC services, management of local infrastructure, website updates, software licensing, hardware procurement, and inventory. I have experience hosting websites (static and server based), as well as benchmarking GPU hardware for scientific applications.

Overall, I believe my combination of research experience, computational skills, interest in applied AI, and willingness to support shared infrastructure would allow me to contribute effectively to the AICH group. I would be grateful for the opportunity to hear more about how I can assist your research.

Sincerely,

ERIC BOITTIER, PH.D.

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ARLESHEIM, BL 4144, SWITZERLAND

Post Doctoral Scientist in Artificial Intelligence building Machine Learning Models and Scientific Software Solutions for Research in the Natural Sciences.

EDUCATION

Doctor of Philosophy, Chemistry

University of Basel

Basel, CH

June 2025

Thesis: *Molecular Deep Learning for Quantitative Simulations and Electrostatic Models.*

Bachelor of Advanced Science (Honours I), Chemistry

The University of Queensland

Brisbane, AU

January 2020

Thesis: *Development of Computational Tools for the Rational Design of Glycosaminoglycan Mimetics.*

RESEARCH EXPERIENCE

Researcher

University of Basel – Department of Chemistry

Basel, CH

January 2020 - Present

- Deep learning for graph-structured scientific data and computational chemistry.
- Design, training, and validation of machine-learned potentials and distributed charge models.
- Top performance in the HyDRA computational vibrational spectroscopy challenge (Physical Chemistry Chemical Physics, 2023).

Research Assistant

Translational Research Institute, Cancer and Aging Research Program

Brisbane, AU

January 2019 - December 2020

- Interdisciplinary collaboration on small-molecule and biomolecular modelling (Schrödinger suite, OpenMM).
- Structure- and ligand-based drug design; data pipelines in Python and R.

INVITED PRESENTATIONS

1. **Eric Boittier.** Bridging machine learning force fields with anisotropic electrostatic models. *Swiss Chemical Society, Machine Learning Group*, Lausanne, CH; Talk.
2. **Eric Boittier.** Mixing machine-learned and empirical energy functions. *apoCHARMM Meeting, NIH*, Boston, US (online); Talk.

SERVICE

Journal of Chemical Physics (AIP Publishing), **Peer reviewer**

Manuscript review

January 2020 - Present

SKILLS

Programming: Python | Julia | R | Fortran | C++ | OpenMPI | CUDA

Frontend Development: Typescript | JavaScript | React | Tailwind CSS

Machine Learning: JAX | PyTorch | TensorFlow | scikit-learn | Weights & Biases

Engineering: Git | CI/CD | Docker | Slurm

Data Science: SnakeMake | SQL | Polars | Pandas

Large Language Models: Unsloth | Hugging Face | LiteLLM | LangChain | OpenClaw

PUBLICATIONS

1. JingChun Wang, Meenu Upadhyay, **Eric D Boittier**, Kham Lek Chaton, Valerii Andreichev, Mike Devereux, Raidel Martin Barrios, Shimoni Patel, Sena Aydin, Kai Töpfer. Cluster Models for Next-Generation, Machine-Learning-Based Energy Functions for Molecular Simulations. *Small Structures* **2026**, 7 (2), e202500656.
2. **Eric D Boittier**, Markus Meuwly. Efficient, Equivariant Predictions of Distributed Charge Models. . Preprint available on *arXiv*, January 01, 2026. DOI: 10.48550/arXiv.2602.07234
3. Kham Lek Chaton, **Eric D Boittier**, Mike Devereux, Markus Meuwly. Explicit, Machine-Learned Two-Body Potentials for Molecular Simulations. . Preprint available on *arXiv*, January 01, 2026. DOI: 10.48550/arXiv.2603.14466
4. **Eric D Boittier**, Silvan Kaser, Markus Meuwly. Roadmap to CCSD (T)-Quality Machine-Learned Potentials for Condensed Phase Simulations. *Journal of Chemical Theory and Computation* **2025**, 21 (18), 8683-8698.
5. **Eric D Boittier**, Silvan Käser, Markus Meuwly. Towards Large-Scale Condensed Phase Simulations using Machine Learned Energy Functions. . Preprint available on *arXiv*, January 01, 2025. DOI: 10.48550/arXiv.2506.23272
6. Wonmuk Hwang, Steven L Austin, Arnaud Blondel, **Eric D Boittier**, Stefan Boresch, Matthias Buck, Joshua Buckner, Amedeo Caflisch, Hao-Ting Chang, Xi Cheng. CHARMM at 45: Enhancements in accessibility, functionality, and speed. *The Journal of Physical Chemistry B* **2024**, 128 (41), 9976-10042.
7. Kai Topfer, **Eric Boittier**, Mike Devereux, Andrea Pasti, Peter Hamm, Markus Meuwly. Force fields for deep eutectic mixtures: application to structure, thermodynamics and 2D-Infrared spectroscopy. *The Journal of Physical Chemistry B* **2024**, 128 (44), 10937-10949.
8. **Eric Boittier**, Kai Töpfer, Mike Devereux, Markus Meuwly. Kernel-based minimal distributed charges: A conformationally dependent ESP-model for molecular simulations. *Journal of Chemical Theory and Computation* **2024**, 20 (18), 8088-8099.
9. Mike Devereux, **Eric D Boittier**, Markus Meuwly. Systematic improvement of empirical energy functions in the era of machine learning. *Journal of Computational Chemistry* **2024**, 45 (22), 1899-1913.
10. Haydar Taylan Turan, **Eric Boittier**, Markus Meuwly. Interaction at a distance: Xenon migration in Mb. *The Journal of Chemical Physics* **2023**, 158 (12).
11. Tabassum Khair Barbhuiya, Mark Fisher, **Eric D Boittier**, Emma Bolderson, Kenneth J O'Byrne, Derek J Richard, Mark Nathaniel Adams, Neha S Gandhi. Structural investigation of CDCA3-Cdh1 p rotein–protein interactions using in vitro studies and molecular dynamics simulation. *Protein Science* **2023**, 32 (3), e4572.
12. Taija L Fischer, Margarethe Bödecker, Sophie M Schweer, Jennifer Dupont, Valéria Lepère, Anne Zehnacker-Rentien, Martin A Suhm, Benjamin Schröder, Tobias Henkes, Diego M Andrada. The first HyDRA challenge for computational vibrational spectroscopy. *Physical Chemistry Chemical Physics* **2023**, 25 (33), 22089-22102.
13. **Eric D Boittier**, Mike Devereux, Markus Meuwly. Molecular dynamics with conformationally dependent, distributed charges. *Journal of Chemical Theory and Computation* **2022**, 18 (12), 7544-7554.
14. **Eric D Boittier**, Norbert Wimmer, Alexandria K Harris, Mark A Schembri, Vito Ferro. Synthesis of a Gal-β-(1? 4)-Gal disaccharide as a ligand for the fimbrial adhesin UcaD. *Australian Journal of Chemistry* **2022**, 76 (1), 30-36.
15. Luis Itza Vazquez-Salazar, **Eric D Boittier**, Markus Meuwly. Uncertainty quantification for predictions of atomistic neural networks. *Chemical science* **2022**, 13 (44), 13068-13084.
16. Katrina Kilday, Neha S Gandhi, Katherine B Sahin, Esha T Shah, **Eric Boittier**, Pascal HG Duijf, Christopher Molloy, Joshua T Burgess, Sam Beard, Emma Bolderson. Elevating CDCA3 levels in non-small cell lung cancer enhances sensitivity to platinum-based chemotherapy. *Communications Biology* **2021**, 4 (1), 638.
17. Sarah-Louise Ryan, Keyur A Dave, Sam Beard, Martina Gyimesi, Matthew McTaggart, Katherine B Sahin, Christopher Molloy, Neha S Gandhi, **Eric Boittier**, Connor G O'Leary. Identification of proteins deregulated by platinum-based chemotherapy as novel biomarkers and therapeutic targets in non-small cell lung cancer. *Frontiers in oncology* **2021**, 11, 615967.

18. Luis Itza Vazquez-Salazar, **Eric D Boittier**, Oliver T Unke, Markus Meuwly. Impact of the characteristics of quantum chemical databases on machine learning prediction of tautomerization energies. *Journal of chemical theory and computation* **2021**, 17 (8), 4769-4785.
19. Silvan Kaser, **Eric D Boittier**, Meenu Upadhyay, Markus Meuwly. Transfer learning to CCSD (T): Accurate anharmonic frequencies from machine learning models. *Journal of Chemical Theory and Computation* **2021**, 17 (6), 3687-3699.
20. **Eric D Boittier**, Yat Yin Tang, McKenna E Buckley, Zachariah P Schuurs, Derek J Richard, Neha S Gandhi. Assessing molecular docking tools to guide targeted drug discovery of CD38 inhibitors. *International journal of molecular sciences* **2020**, 21 (15), 5183.
21. **Eric D Boittier**, Jed M Burns, Neha S Gandhi, Vito Ferro. GlycoTorch Vina: docking designed and tested for glycosaminoglycans. *Journal of Chemical Information and Modeling* **2020**, 60 (12), 6328-6343.
22. **Eric D Boittier**, Neha S Gandhi, Vito Ferro, Deirdre R Coombe. Cross-species analysis of glycosaminoglycan binding proteins reveals some animal models are “More Equal” than others. *Molecules* **2019**, 24 (5), 924.
23. Jed M Burns, **Eric D Boittier**. Pathway Bifurcation in the (4 + 3)/(5 + 2)-Cycloaddition of Butadiene and Oxidopyrylium Ylides: The Significance of Molecular Orbital Isosymmetry.. *Journal of Organic Chemistry* **2019**, 84 (10), 5997-6005.

PROFESSIONAL REFERENCES

The following academics can speak to my doctoral research in computational chemistry, machine learning, and scientific software at the University of Basel.

Dr. Michael Devereux

Scientific Staff/Research IT, Computational Physical Chemistry
Department of Chemistry, University of Basel
Klingelbergstrasse 80, CH-4056 Basel, Switzerland
Email: michael.devereux@unibas.ch
Telephone: +41 61 207 38 20

Prof. Dr. Markus Meuwly

Professor of Physical Chemistry; Group Leader (Group Meuwly)
Department of Chemistry, University of Basel
Klingelbergstrasse 80, CH-4056 Basel, Switzerland
Email: m.meuwly@unibas.ch
Telephone: +41 61 207 38 21

ONLINE PORTFOLIOS AND TECHNICAL WORK



Google Scholar

Scholar profile (publications & citations)



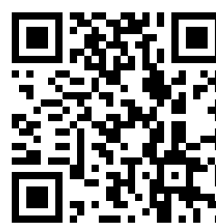
GitHub

github.com/EricBoittier



Personal website

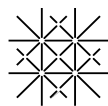
ericboittier.github.io



Hugging Face

huggingface.co/EricBoi

These profiles summarize publications — along with code, notes, demos, datasets, and other technical artefacts from my research.



Universität
Basel

Philosophisch-Naturwissenschaftliche
Fakultät



ERIC DAVID BOITTIER

geboren am 11. Januar 1997,

hat am 15. Januar 2026 das

DOKTORATSEXAMEN

im Promotionsfach

CHEMIE

bestanden.

Die Promotionsurkunde wird ausgehändigt, sobald die Dissertation

„Elements of molecular deep learning for quantitative simulations & electrostatic modelling. Scale, symmetry, synthesis”

gemäss den Bestimmungen über die Ablieferung der Pflichtexemplare bei der Universitätsbibliothek Basel eingereicht wurde.

Bis dahin trägt der Promovierte den Titel «Dr. phil. designatus» (Dr. phil. des.).

Basel, 15. Januar 2026

Prof. Dr. Heiko Schuldt
Dekan

ERIC DAVID BOITTIER (43908617)

Page 1

AWARDS

BACHELOR OF ADVANCED SCIENCE (HONOURS) WITH HONOURS CLASS I WITH A MAJOR IN CHEMISTRY

AWARD: BACHELOR OF ADVANCED SCIENCE (HONOURS) (AQF:8)
LEVEL: HONOURS CLASS 1
DATE OF CONFERRAL: 18 JULY 2019

COURSE	#	GRADE
ENROLMENT HISTORY DETAILS		
UNDERGRADUATE		
SEMESTER 1, 2015		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (AQF: 8)		
BIOL 1020	GENES, CELLS & EVOLUTION	2.00 5
CHEM 1100	CHEMISTRY 1	2.00 6
MATH 1050	MATHEMATICAL FOUNDATIONS	2.00 5
SCIE 1100	ADV THEORY AND PRACTICE	2.00 7
		SEM GPA: 5.75
SEMESTER 2, 2015		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (AQF: 8)		
CHEM 1200	CHEMISTRY 2	2.00 6
INFS 1200	INTRO TO INFORMATION SYSTEMS	2.00 6
SCIE 2111	INTRO TO SCIENCE RESEARCH	2.00 7
STAT 1301	ADV ANALYSIS SCIENTIFIC DATA	2.00 6
		SEM GPA: 6.25
SEMESTER 1, 2016		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (AQF: 8)		
CHEM 2050	ORGANIC & INORGANIC CHEMISTRY	2.00 4
CHEM 2054	EXPERIMENTAL CHEMISTRY	2.00 6
CSSE 1001	SOFTWARE ENGINEERING	2.00 6
MATH 1061	DISCRETE MATHEMATICS	2.00 4
		SEM GPA: 5.00
SEMESTER 2, 2016		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (AQF: 8)		
CHEM 2002	PHYSICAL CHEMISTRY	2.00 5
CHEM 3008	MEDICINAL & BIOL CHEMISTY	2.00 5
SCIE 3260	RESEARCH-CHEM, BIOC, MICR (A)	2.00 7
		SEM GPA: 5.67
SEMESTER 1, 2017		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (Chemistry) (AQF: 8)		
CHEM 3001	ADVANCED ORGANIC CHEMISTRY	2.00 5
CHEM 3004	DETERMINAT OF MOLECULAR STRUC	2.00 5
CHEM 3010	ADVANCED INORGANIC CHEMISTRY	2.00 5
CHEM 3013	NANOSCIENCE: SELF-ASSEMBLY	2.00 5
		SEM GPA: 5.00
SEMESTER 2, 2017		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (Chemistry) (AQF: 8)		
CHEM 3011	ADVANCED PHYSICAL CHEMISTRY	2.00 6
CHEM 3016	ADVANCED EXPERIMENTAL CHEM	2.00 6
MATH 1051	CALCULUS & LINEAR ALGEBRA I	2.00 5
		SEM GPA: 5.67

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Continued on page 2

ERIC DAVID BOITTIER (43908617)

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COURSE	#	GRADE
SEMESTER 1, 2018		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (Chemistry) (AQF: 8)		
CSSE 2002	PROGRAMMING IN THE LARGE	2.00 5
SCIE 3261	RESEARCH-CHEM, BIOC, MICR (B)	2.00 6
SEM GPA: 5.50		
SEMESTER 2, 2018		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (Chemistry) (AQF: 8)		
CHEM 6512	HONOURS IN CHEMISTRY	0.00 IP
SEM GPA: 0.00		
SEMESTER 1, 2019		
BACHELOR OF ADVANCED SCIENCE (HONOURS) (Chemistry) (AQF: 8)		
CHEM 6512	HONOURS IN CHEMISTRY	16.00 7
SEM GPA: 7.00		
Dean's Commendation for Academic Excellence		

Entries below this line are invalid



Mark Erickson
Academic Registrar
18 JULY 2019

For students who began their programs in 1974 or later, computer generated academic records replaced those issued earlier. Where a student first enrolled prior to 1974, the entry “credit gained before 1974” indicates a full academic record is shown in two separate statements. From 1990 the Queensland Agricultural College (QAC) became The University of Queensland Gatton College and is now designated University of Queensland, Gatton. Students enrolled at QAC prior to 1990 who then enrolled at the University will have a separate statement for their pre-1990 studies.

The Academic Year

Programs are offered across a range of formats: semesters, trimesters, teaching periods and research quarters which span the calendar year. Details of the academic calendar can be found on The University of Queensland website.

The Grading System

The University of Queensland grading system may be found at: <http://ppl.app.uq.edu.au/content/3.10.07-grading-system>

Grade	Description
7	High Distinction
6	Distinction
5	Credit
4	Pass
3	Fail ¹
2	Fail
1	Fail
P	Non-graded pass
N	Non-graded fail

Other symbols used, where a result has not been awarded, may include:

—	Incomplete result (pre-2001)
AT	Attended lectures
EX	Exempt from course
G	Credit for the course cancelled or refused
INC	Incomplete result (2001 onwards)
IPR	In progress
K	Late cancellation
M	Senate determined that credit for course be cancelled or refused
Q	Qualifying examination (P = passed, N = failed)
RW	Result withheld
S	Supplementary assessment granted
SP	Special/Deferred examination granted
SR	Course retaken
T	Transfer credit/Advanced Standing
THS	Thesis submitted
X	No assessable work received

¹ From semester 1 2006, grade of 3 is a failing grade

Notings:

Symbols

#	represents the credit point value (pre-2001) or unit value (2001 onwards) of a course.
GPA	stands for Grade Point Average.

Dean's Commendation

A Dean's Commendation for Academic Excellence recognises students who have excelled academically. The criteria for Dean's Commendation for Academic Excellence may be found at: <https://ppl.app.uq.edu.au/content/3.50.04-deans-commendations-academic-excellence>

Honours

Honours are awarded in one of three classes: honours class I; honours class II (A or B); honours class III (A or B). From Semester 2, 2005, the use of weighted percentage scores to calculate the class of honours, using University-wide cutoff scores, was introduced progressively across all honours programs. From semester 1, 2015 the use of a weighted GPA to calculate the honours class was introduced progressively across all honours programs. The criteria for the award of honours may be found at: <https://ppl.app.uq.edu.au/content/3.50.12-bachelor-honours-degrees>